MD. ABU SUFYAN

Software Engineering | Applied AI Research | Technical Education

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PROFILE

Junior Software Engineer at Scalius Labs, AI and machine learning instructor, and RUET CSE graduate. Co-author of two ICCiT 2024 papers, with project experience in web APIs, computer vision and explainable intrusion detection. Interested in trustworthy AI, practical software engineering and research collaboration.

EDUCATION

B.Sc. in Computer Science & Engineering | RUET

Graduation year: 2024 | CGPA: 3.60/4.00 | Academic session: 2020–2021
Final result published: 11 August 2026

HSC: Gurudoyal Govt College, 2020 | SSC: Alhaj Amir Uddin High School, 2018

EXPERIENCE

Junior Software Engineer | Scalius Labs

September 2026 – Present

Software engineering using Codex and Claude as AI-assisted development tools.

Research Assistant | Machine Learning Research Group, RUET

July 2024 – Present

Developed and evaluated NLP and forecasting models; contributed to preprocessing, experiments, analysis and technical writing for two ICCiT 2024 papers.

Machine Learning & AI Instructor | Artificial Intelligence Bangladesh

March 2024 – Present

Delivered Python, machine learning, deep learning and computer vision instruction, with practical materials spanning data preparation, training and evaluation.

Python Trainer | Bangladesh Computer Council, Rajshahi

2026

Conducted Python instruction and mentoring through practical programming exercises and debugging.

Industrial Trainee | BrainStation 23 PLC

March 2025 | 15 days

Contributed to a live-video traffic-sign detection web application using YOLOv8, MobileNet and ResNet, including training and frontend/backend integration in a team workflow.

TECHNICAL SKILLS

Programming: Python, Go, JavaScript, SQL, C/C++, PHP, Dart.
AI and data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, BiLSTM, XGBoost, Prophet, computer vision, SHAP, IBM ART.
Applications: React, Node.js, Express, Laravel, Echo, GORM, PostgreSQL, MySQL, MongoDB, Flutter, Riverpod, Firebase.
Tools and foundations: Git, Linux, Docker, Postman; foundational Verilog and introductory SystemVerilog.
Currently exploring: LLMs, RAG, LangChain, LangGraph, vector databases and FastAPI.

RESEARCH, PROJECTS & COMMUNITY

SELECTED PROJECTS

Explainable Network Intrusion Detection | Undergraduate thesis

Thesis project

CNN-BiLSTM detector; FGSM, PGD and C&W adversarial evaluation through IBM ART; SHAP interpretation and an interactive defense visualizer.

SpotSync | Smart Parking & EV Charging REST API

Academic project

Go, Echo v4, GORM and PostgreSQL backend with STRIDE/DREAD threat modelling as a security coursework case study.

Attendance Management App | Mobile application

2024

Flutter, Dart, Firebase authentication and real-time database, Riverpod, course selection and attendance history.

Project NEXUS | RUET IoT Club

2025

Water-quality sensing, embedded prototyping and data acquisition.

Additional work: Traffic-sign detection, Gemini Health AI prototype, full-stack web applications and an in-development retrieval-based medical chatbot. EcoHomeAI is an early-stage smart-home venture; no commercial results claimed.

PUBLICATIONS

Co-author, 27th IEEE ICCiT 2024, Cox’s Bazar, Bangladesh, 20–22 December 2024:

1. **A Cross-Analyzing Approach to Sentiment and Bias Detection in Social Media: Insights from Geopolitical Conflicts.** Compared BERT and BiLSTM approaches with WordPiece and GloVe representations.

2. **Improving Monkeypox Outbreak Prediction Through Time-Series Forecasting with Machine Learning Models.** Compared XGBoost, LSTM and Prophet on monkeypox death time-series data.

LEADERSHIP & SERVICE

RUET IoT Club: Alumni Affairs Advisor, June 2026–Present; former President, February 2025–June 2026; previously Joint Organizing Secretary.
RUET Computing Society: Alumni member from June 2026; former President and Founding Member, January–June 2026.
FutureNation: Campus Representative, June 2025–Present; Campus Volunteer Recognition, February 2026.
VBD, Rajshahi: HR Officer, January–December 2025; appreciation issued 1 January 2026.
Aachol Foundation: Former HR Executive, July 2025–June 2026.
JAAGO: Youth trainer, Ekannor Jaagoron divisional session, December 2025.
Campus outreach: Programming Hero, weDevs Academy, WSDA New Zealand and ICSC ambassador roles; dates not stated in source CVs. Other service includes class representation and community volunteering.

SELECTED TRAINING & RECOGNITION

Duke/Coursera Data Science Math Skills (2025); Kaggle Intro to ML (2024) and Deep Learning (2025); Google Generative AI for Educators (2025); Anthropic Claude 101 (date unstated); EDGE PHP/Laravel, 80 hours (2025); Ostad Flutter (date unstated); UNDP SDG Primer (2025) and Environmental Stewardship (2026); Public Policy 101 (2026). Simplilearn CAPM and Ostad CCNA-related records are course completions, not PMI/Cisco professional certifications.

11th Bangladesh Chemistry Olympiad 2020 preliminary round: joint 22nd (19) position; event held 27 August 2021.

Languages: Bangla (native), English (professional working proficiency).